Correct before the deadline - or NONE?

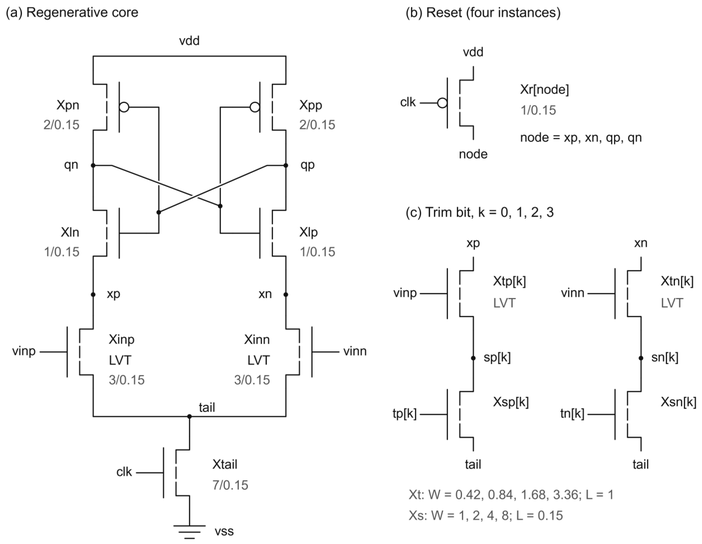
Comparator Atlas | SKY130 StrongARM comparators

Wei-Lun Hsu | National Tsing Hua University

IEEE SSCS Code-a-Chip / ISSCC 2027

1. Define a correct decision

Calibration is not a timing guarantee.
Every included sample must be correct on time.



W/L in µm. All nMOS bulks: vss; pMOS bulks: vdd. Bulk wires omitted.

Established StrongARM topology in SKY130. Nine sizing candidates; the lower-energy control was tested after selection.

Correct, wrong or unresolved

Correct: complementary 80% / 20% output rails, expected polarity, before the deadline. Wrong polarity and unresolved outputs are separate outcomes.

Read stored measurements, not new simulations. A later deadline changes the readout, not the full 20-30 ns core-energy cycle.

Later extension (Notebook Section 9; not part of this sealed record): unit6_w6, a 6-bit unit trim with input W = 6 um, qualifies at +/-1 mV at 1 and 2 ns over all 49 schematic conditions, at 760 fJ mean core energy. No layout of its own.

2. Qualify before ranking energy

Calibrated SCHEMATIC, not layout RC.
local_boundary; all 49 controlled-width-stress conditions.
Both signed samples through 30 mV; 1 ns deadline.

>= 1 mV **NONE**

No feasible compared design

Mean energy / limits: not defined

>= 3 mV **Selected: lvt_balanced_4b**

294/294 correct at all included samples

249.655 fJ mean core / cycle

>= 30 mV **Control: lvt_base_3b**

98/98 correct at all included samples

150.531 fJ mean core / cycle

Post-hoc comparison of three designs. Different bands are different specs, not energy savings. No interpolation or global optimum.

Selected >= 3 mV band / same 294 samples

154.517 ps

Min. deadline - recorded decision time (sampled)

363.217 fJ

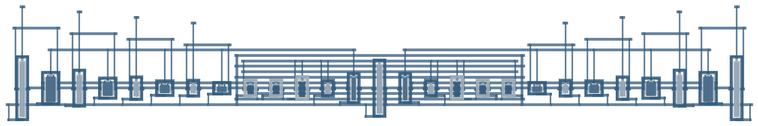
Maximum sampled core energy

20 wrong samples remain at 2 ns

Exactly the same keyed set at 1 ns and 2 ns, all at signed +/-1 mV. Locations and trim codes do not identify a physical failure cause.

3. Separate layout evidence

Local-pitch b1 vs adopted a1: nominal, code-zero, known nonblind. 45 PVT conditions x signed +/-3 and +/-10 mV. 180 keys per layout / C or RC; ngspice-47.

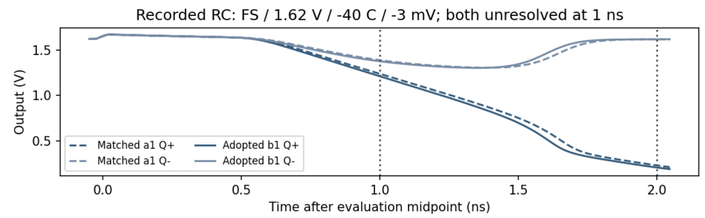


Bbox (um2): adopted 1873.300 / matched a1 2074.254 (9.69% less).
Historical native proofs reused, not rerun; no signoff.

Correct / 180	1 ns	2 ns
Both C	170/180	180/180
a1 RC	156/180	180/180
b1 RC	158/180	180/180

RC mean: 421.04 -> 412.88 fJ / cycle

1 ns RC: 24 -> 22 unresolved; zero wrong.
Only SS 1.62 V / 125 C / +/-10 mV improve; FS unresolved.
2 ns declared primary; no lost correct keys.



Matched a1 -> b1; worst RC: 1.811 -> 1.760 ns.
~1.94% energy change is not robust global PPA.

Archived RC-deck outcomes; model physical fidelity not yet qualified.

Source Notebook Source Colab

No qualified PEX or silicon prediction. C-only is not independent ground truth; RC-versus-C differences do not isolate resistance.

Finite samples, not foundry Monte Carlo/yield, noise/jitter/PVT confidence, timing signoff or a worst-cycle guarantee. Core excludes drivers and calibration/controller infrastructure.

References: Razavi (2015), DOI 10.1109/MSSC.2015.2418155; Li, Xu & Iizuka (2022), DOI 10.1007/s10470-022-01992-6.

Original code: MIT; SKY130: Apache-2.0. GitHub Copilot assisted implementation, experiment automation, figures and documentation; the author is responsible.